

Measure-to-measure interpolation using Transformers: formalization

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0.1 Foundations

Definition 1 (Tangential projector). The layer-normalization projector $P_x^\perp v = v - \langle x, v \rangle x$ onto the tangent space.

Lemma 2 (Projector identity, L1). For $\|x\| = 1$, $P_x^\perp$ is symmetric and idempotent, $P_x^\perp x = 0$, and $\langle P_x^\perp v, v \rangle = \|v\|^2 - \langle x, v \rangle^2$. [machine-checked]

Definition 3 (Geodesic distance). $d_g(x, y) = \arccos \langle x, y \rangle$, with $\cos d_g(x, y) = \langle x, y \rangle$ for unit vectors. [machine-checked]

0.2 Labeled axioms (missing Mathlib infrastructure)

Definition 4 (Wasserstein distance). W_2, W_1 with nonnegativity, symmetry, triangle inequality, the map-coupling bound, and Kantorovich-Rubinstein duality. [axiomatised]

Definition 5 (Continuity-equation flow). The solution map Φ_θ^t on points and on measures, with Lipschitz, bijective and parked-region-identity properties; the switch count. [axiomatised]

0.3 Self-contained leaves

Lemma 6 (Gate algebra and gate ODE, L2). $\frac{d}{dt} \langle x(t), \omega \rangle = (\cos R - \langle z, x \rangle)_+ (1 - \langle x, \omega \rangle^2)$ along the gated flow, with the gate active exactly on $\{d_g(z, x) > R\}$ (eq. B.4-B.5). [machine-checked] The kernel-checked equivalence $g(x) > 0 \iff d_g(z, x) > R$ corrects the printed (B.4), which states the gate is active on $\mathcal{B}_0 = B(z, R)$; consistency of Lemma B.2 requires the opposite sign (finding F1 in *RESEARCH.md*, confirmed independently by experiment E1).

Lemma 7 (Barycenter ODE, L6). $\frac{d}{dt} \langle x, \alpha \rangle = c (1 - \langle x, \alpha \rangle^2)$ and strict increase for $c > 0$ (eq. B.9). [machine-checked]

Lemma 8 (Geodesic gradient, L4). $\frac{d}{dt} d_g(x(t), \omega) = -(1 - \langle x, \omega \rangle^2)^{-1/2} \langle x', \omega \rangle$, exhibiting $-P_x^\perp \omega / \sqrt{1 - \langle x, \omega \rangle^2}$ as the Riemannian gradient (4.4). [machine-checked]

Lemma 9 (Separating hyperplane, L3). $d_g(\omega, x) \geq \pi/2$ and $\tau \in (0, 3\pi/8)$ give $\langle \omega, x \rangle < \cos(\pi/8 + \tau)$ (Prop 4.2 Step 1). [machine-checked]

Lemma 10 (Lyapunov sign, L5). $E = 1 - \cos \theta$ along $\theta' = -\alpha \sin \theta$ has $\dot{E} = -\alpha \sin^2 \theta \leq 0$ (Ex. 6.1). [machine-checked]

Lemma 11 (Ball-chain retention, L9). Per-step retention $(1 - \varepsilon)$ gives $a_K \geq (1 - \varepsilon)^K a_0$ (Lemma B.1 induction). [machine-checked]

Lemma 12 (Pigeonhole, L10). A nonempty open ball contains a point $\neq a$, so no map is constant on it (Lemma 3.4 Part 1). [machine-checked]

Lemma 13 (Disjoint hulls give non-colinear barycenters, L11). Modeling $\text{conv}_g(s) = \text{cone}(s) \cap \mathbb{S}^{d-1}$ (faithful in a hemisphere), empirical measures in the orthant with disjoint geodesic hulls have non-colinear barycenters (not SameRay). Discharges the step Proposition 3.1's induction asserts without proof (finding F2); kernel-clean for the empirical regime of Thm 1.1 / restricted Thm 1.2. [machine-checked]

Lemma 14 (F2, general probability measures, L11'). *For arbitrary probability measures supported in disjoint closed convex cones the barycenters are non-colinear, via Mathlib's `Convex.integral_mem` (the barycenter lies in any closed convex set carrying full mass). Closes the general-measure residual of F2 with no new axioms. [machine-checked]*

Lemma 15 (Linearized OT bound, L7). $W_2(T_{\#}^1\mu, T_{\#}^2\mu) \leq \|T^1 - T^2\|_{L^2(\mu)}$ (Lemma 5.2). [axiomatised]

Lemma 16 (Markov bound, L8). $W_2(\mu, \delta_{x_0}) \leq \eta_2 \Rightarrow 1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ (Claim 2). *The truncated-distance bump is machine-checked 1-Lipschitz; the Markov inequality is derived from it via W_1 Kantorovich-Rubinstein duality. [axiomatised]*

0.4 Mass transport and clustering

Lemma 17 (Single ball pair, B.2). $\text{Mass } \mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1 - \varepsilon)\mu_0(\mathcal{B}_0)$. [open] *Caveat (F1): the printed parameters $U = -z\mathbf{1}^\top$, $b = \cos(R)\mathbf{1}$ make the gate active on the complement of $\mathcal{B}_0$, so the lemma transports nothing as written; the sign must be flipped to $U = +z\mathbf{1}^\top$, $b = -\cos(R)\mathbf{1}$. True once corrected.*

Lemma 18 (Ball chain, B.1). $\text{Mass through } K \text{ overlapping balls} \geq (1 - \varepsilon)^K$. [open]

Proposition 19 (Clustering to a point, 2.1). *A single-hemisphere measure clusters to a point; $\text{diam} \rightarrow 0$ at rate $O(\log 1/\varepsilon)$. [open]*

Proposition 20 (Clustering to discrete, 2.2). *Disjoint-support atomless measures cluster to M atoms. [open]*

0.5 Disentangling supports

Lemma 21 (Transport to the orthant, 3.2). *One switch moves measures into $\mathbb{Q}_1^{d-1}$. [open]*

Lemma 22 (Disentangle non-colinear, 3.3). *Shrink a measure's hull to its barycenter direction. [open]*

Lemma 23 (Make non-colinear, 3.4). *Two measures made non-colinear. [open]*

Proposition 24 (Disentanglement, 3.1). *A piecewise-constant θ renders the N supports pairwise disjoint, $O(dN)$ switches. [open] The "disjoint hulls $\Rightarrow$ non-colinear barycenters" step the induction asserted without proof (finding F2) is now the machine-checked leaf L11 (13) for the empirical regime; the headline stays open only because it rests on the flow / conv_g -nesting axioms.*

0.6 Matching discrete measures

Proposition 25 (Steer one point, 4.2). $x_0^M \rightarrow y^M$ while inactive points stay fixed, ≤ 6 switches. [open]

Proposition 26 (Match ensembles, 4.1). M points $x_0^i \rightarrow y^i$ exactly, $\leq 6M$ switches. [open]

0.7 Main results

Lemma 27 (Transport maps preserved, 5.1). *Existence of ψ . [open]*

Lemma 28 (L^2 approximation, 5.4). *Approximate any ψ by a flow map. [open]*

Theorem 29 (Theorem 1.2). *A single Transformer matches N arbitrary inputs to N arbitrary (transport-matchable) targets in W_2 . [open]*

Theorem 30 (Theorem 1.1). *Dirac targets: matching with $O(dN)$ switches and norm $O(dN/T + \log 1/\varepsilon)$. [open]*

Proposition 31 (Generic $O(1)$ switches, 6.2). *For generic i.i.d. discrete inputs a single constant parameter disentangles almost surely. [informal]*